

Work Instructions

BT AS-ILT015, TD10006353 EN08

Scope of validity	AS-LT
Application	Comissioning of labelling technology - Check of the safety and EMERGENCY STOP functions for safety switching devices (PNOZsigma)

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Caution!

These instructions are designed to test the safety technology with safety switching devices starting from series 2009/04. The tests should take place at the beginning of commissioning to guarantee functional safety technology for continued work on the machine. Everyone taking part in the test must have been trained for testing safety technology! Necessary measurements must be performed by an electrician!

Work Instructions**BT AS-ILT015, TD10006353 EN08****History of documents**

Revision	Datum	Kommentar
01 → 02	30.06.2010	Scope of validity / Application to MO-SET labelling technology changed 1.3 Description of the safe outputs and the check-back signal contacts SO1 and SO2 Description justified 5.4 label carriage forward new revised (Anton Gareis)
02 → 03	14.09.2010	1.3 Description of the outputs and the LED displays SO8 (Enable circuit 8) from Guards overall delayed to Guards overall without delay justified 3.4 LED In2 at the Contact extension device justified (Anton Gareis)
03 → 04	24.01.2012	5.4 label carriage forward new revised; 5.1 machine conveyor run if guard is opened (Josef Raith)
04 → 05	24.08.2012	Designation certifying circle changed (Josef Raith)
05 → 06	17.10.2012	1.3 picture for S9 added 2.1 Pilz PNOZsigma S9 added 5.3 „Press an EMERGENCY STOP button“ deleted (Andreas Reichinger)
06 → 07	05.08.2012	3.4 connecting and disconnecting S12 for non-contact-switches added (Andreas Reichinger)
07 → 08	05.11.2014	New design of the document New name of the department added Names of the tables and figures added. (Andreas Reichinger)

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1 Description

1.1 Introduction

This checklist should be processed before the initial startup of all drives and safety-relevant components to guarantee that the safety functions of the machines are working properly.

With these instructions, safety functions of labelling and monitoring technology created with the "PNOZsigma" safety switching system can be tested.

Basically, such a test of the safety technology is divided into three parts. First of all, a device test of the safety switching devices regarding operational readiness and correct settings is conducted. Then the function of the safety sensors as well as the reaction of the switching devices is tested (safety logic). Finally, a test of the wiring of the safety-related potentials (between the safety logic and the actuators) is performed. This should primarily take place by means of measurements. In addition, defective actuators may hereby be detected.

1.1.1 Prerequisites:

Everyone taking part in the test must have been trained for testing safety technology. We recommend that the acceptance inspection of the safety technology be conducted by two skilled electricians, because, for example, one person can activate the safety sensors while the other person performs measurements and visual inspections in the control cabinet.

The safety devices on the machine to be tested must be installed.

Make sure the machine is empty and the shafts can move freely.

1.1.2 Tools / aids:

Multimeter and electrical tools

1.2 Safety

Before the start of the measurement, other people should be prevented from accessing the machine.

Boundary tape with notices

closed guard doors (would be important; however, normally not possible)

The machine must be idling:

No packs must be in the machine.

The individual components of the respective motors must be able to rotate freely.

If necessary, remove components.

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1.3 Description of the outputs and the LED displays

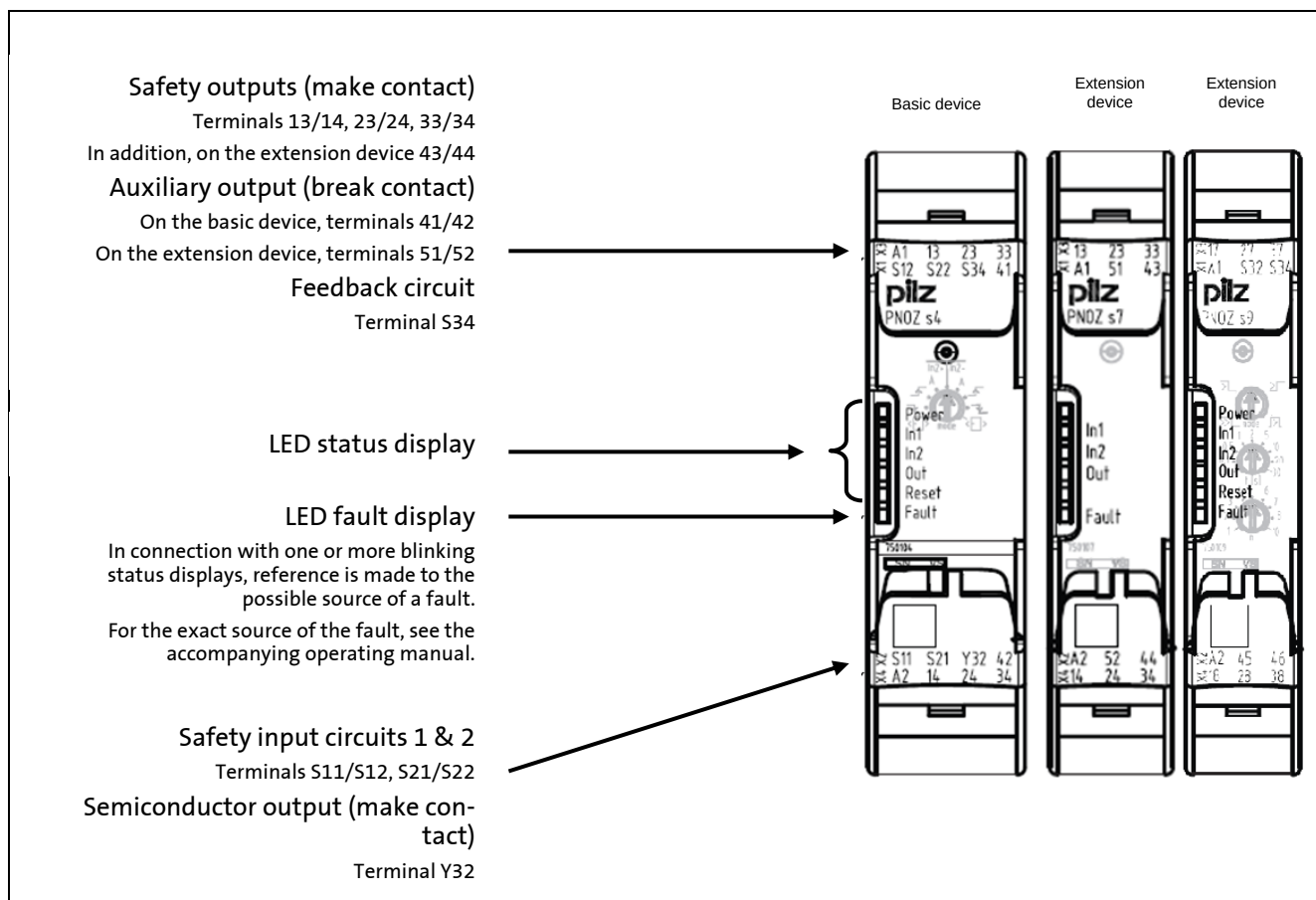


Fig.: 1.3./

LED	Terminal(s)	Function	Description
Power	A1 / A2	Supply voltage	Supply voltage provided
In1	S11 / S12	Safety input circuit 1	Safety-relevant sensors, circuit 1
In2	S21 / S22	Safety input circuit 2	Safety-relevant sensors, circuit 2
Out	13 / 14 23 / 24 33 / 34 43 / 44 (PNOZ s7) Y32	Safety contacts	After the input circuits 1 and 2 are closed, the safety contacts are closed Semiconductor output Y32 carries a high signal In case of deactivation, no delay In case of reactivation, no delay
RESET	S34	RESET / start	24 V DC are applied to the terminal

Tab.: 1.3./

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Output	FBG	LEDs	Function	Description
SO1 (Enable circuit 1)	=.1101	Out	EMERGENCY STOP intrinsic	■ Every EMERGENCY STOP button of the machine is read in order without delay. ■ In addition, EMERGENCY STOP of internal branch subsystems (e.g. Labelinspection, Dating unit, etc.) ■ In case of deactivation, no delay ■ In case of reactivation, no delay
SO2 (Enable circuit 2)	=.1101	Out	EMERGENCY STOP overall	■ Every EMERGENCY STOP button of the machine is read in order without delay. ■ In addition, EMERGENCY STOP of external branch subsystems (e.g. container conveyor, Filling machine, Bridge third-party machine etc.) ■ In case of deactivation, no delay ■ In case of reactivation, no delay
SO3 (Enable circuit 3)	=.1201	Out	Combined safety output for safe stop of the machine / (EMERGENCY STOP delayed, guards with JOG override delayed SLS directly)	■ EMERGENCY STOP button of the machine ■ EMERGENCY STOP button of internal and external subsystems ■ Guard doors of the machine and possibly of subsystems ■ JOG switches and operating mode switches (override certain guard doors - see the connection diagram) ■ Safely-limited speed (SLS) ■ In case of deactivation in production mode: 2.5 seconds delay ■ In case of deactivation in set-up mode: 2.5 seconds delay ■ In case of reactivation: no delay
SO4 (Enable circuit 4)	=.1201	Out	Enable SLS (reduced speed)	■ Machine guard doors delayed with 2.5 seconds
SO5 (Enable circuit 5)	=.1101	Out	EMERGENCY STOP overall delayed	■ Every EMERGENCY STOP button of the machine is read in order without delay. ■ In addition, EMERGENCY STOP of external and internal subsystems ■ In case of deactivation, 2.5 seconds delay ■ In case of reactivation, no delay
SO8 (Enable circuit 8)	=.1201	Out	Guards overall without delay	■ Guard doors of the machine

Tab.: 1.3.II

1.4 Delay time

- Delay time according to the specification for these machines from SIET.

Labelling technology: Delay time: 2.5 s

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2 Device test / safety switching devices "PNOZsigma"

2.1 Test of the set operating mode at the rotary switch

2.1.1 Test procedure

At the start of the test of the safety technology, the position of the rotary switches (mode of operation) is to be visually checked at the devices as specified in the connection diagram. For this, it is not necessary that voltage be applied to the devices.

The following device types (basic devices) can hereby be tested:

- Pilz PNOZsigma S4
- Pilz PNOZsigma S5
- Pilz PNOZsigma S9

2.1.2 Explanation of the standard settings at Krones

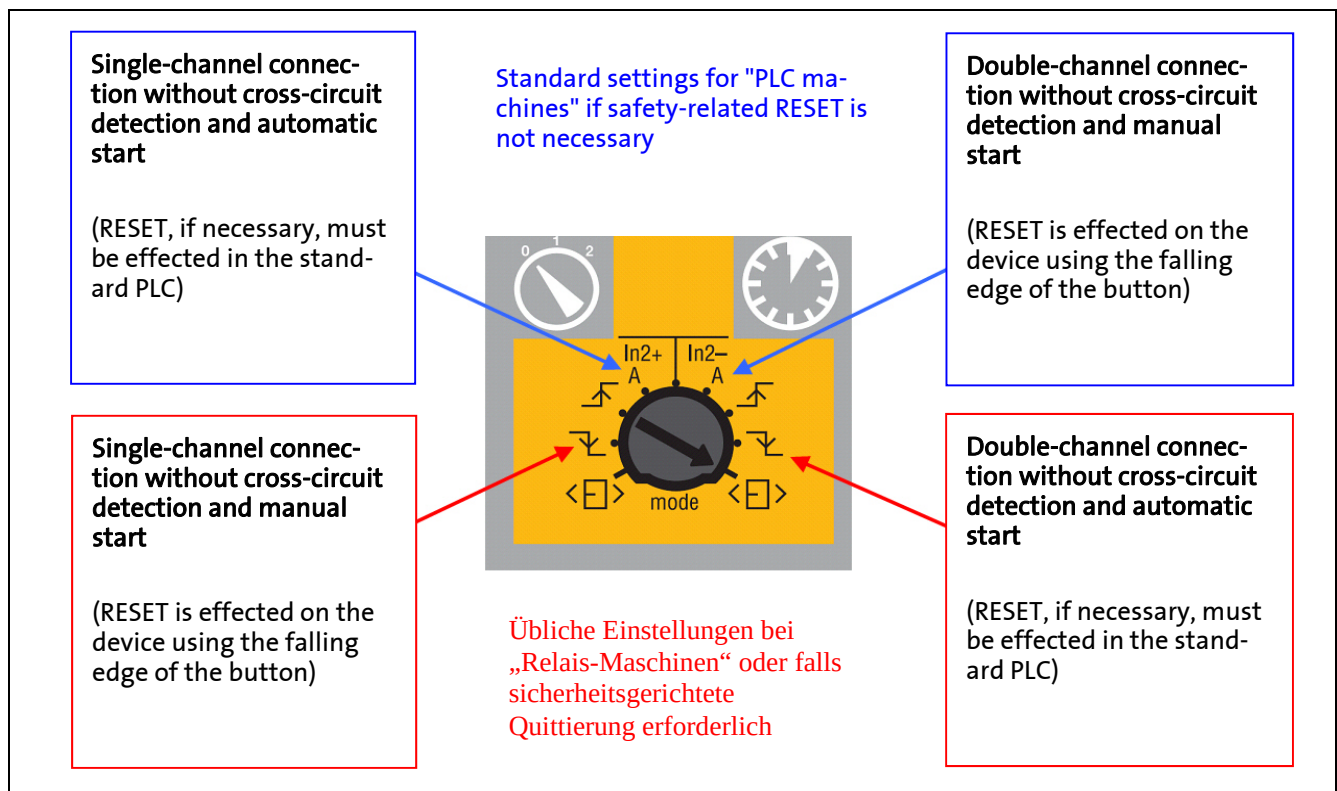


Fig.: 2.1.2./

2.2 Test of the device functions (readiness for operation)

After the power supply is turned ON, the readiness for operation of all the devices is to be visually checked (LEDs on the devices). For this, all existing safety sensors are to be returned to their original state and possibly reset.

Initial situation:

- Machine stopped
- No EMERGENCY STOP switch activated
- Guard doors are closed (if provided)
- Key switch set to "Production" (if provided)
- JOG switch not activated (if provided)

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Device status for selected basic devices (e.g. "intrinsic" EMERGENCY STOP and no EMERGENCY STOP activated):

- LED "Power" → ON (continuous light)
- LED "In1" → ON (continuous light)
- LED "In2" → ON (continuous light)
- LED "Out" → ON (continuous light)
- LED "Reset" → depending on the connection (AUTO start or manual/monitored start)
- LED "Fault" → OFF

Device status for selected contact extension:

(if provided)

- LED "In1" → ON (continuous light)
- LED "In2" → ON (continuous light)
- LED "Out" → ON (continuous light)
- LED "Fault" → OFF

Device status for not selected basic devices (e.g. basic device "JOG switch" and this JOG switch not activated):

(if provided)

- LED "Power" → ON (continuous light)
- LED "In1" → OFF
- LED "In2" → OFF
- LED "Out" → OFF
- LED "RESET" → OFF
- LED "Fault" → OFF

Device status for not selected contact extension:

(if provided)

- LED "In1" → OFF
- LED "In2" → OFF
- LED "Out" → OFF
- LED "Fault" → OFF

Note:

If, in spite of applied power supply, all the LEDs (including "Power") on a basic device are OFF, this is generally not a case of a faulty device. This condition is usually due to a cross circuit between S11 and S21 or between S12 and S22, whereby the voltage in the device collapses.

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3 Test of the EMERGENCY STOP button / SAFETY SWITCH

3.1 Test of all "intrinsic" EMERGENCY STOP switches

- Key switch on production mode / All guard doors closed / No EMERGENCY STOP activated / machine ON
- Each "intrinsic" EMERGENCY STOP switch should be activated one after the other. Check the functions according to the following table.
- All the malfunction and help displays on the touch-screen must be checked. (if provided)

Visual inspection			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
SO1 LED Out	LED turns OFF	LED Out	LED turns OFF
SO2 LED Out	LED turns OFF EMERGENCY STOP of the machine	LED Out	LED turns OFF
SO3 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO4 LED Out	LED ON	LED Out	LED ON
SO5 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO8 LED Out	LED ON	LED Out	LED ON

Tab.: 3.1.I

- Key switch on set-up mode / All guard doors closed / No EMERGENCY STOP activated / Machine stopped
- Each "intrinsic" EMERGENCY STOP switch should be activated one after the other. The functions should be checked according to the following table.
- All the malfunction and help displays on the touch-screen must be checked. (if provided)

Visual inspection			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
SO1 LED Out	LED turns OFF	LED Out	LED turns OFF
SO2 LED Out	LED turns OFF EMERGENCY STOP of the machine	LED Out	LED turns OFF
SO3 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO4 LED Out	LED ON	LED Out	LED ON
SO5 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO8 LED Out	LED ON	LED Out	LED ON

Tab.: 3.1.II

Inputs/outputs which are not used cannot be checked. Which outputs on the respective machine are to be used is to be found in the connection diagram.

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3.2 Test of all "third-party" EMERGENCY STOP signals

- Key switch on production mode / All guard doors closed / No EMERGENCY STOP activated / machine ON
- Each "third-party" EMERGENCY STOP switch should be activated one after the other. Check the functions according to the following table.
- If no "third-party" EMERGENCY STOP mechanism is present, the jumpers for the "third-party" EMERGENCY STOP should be removed.
- All the malfunction and help displays on the touch-screen must be checked. (if provided)

Visual inspection			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
SO1 LED Out	LED ON	LED Out	LED ON
SO2 LED Out	LED turns OFF EMERGENCY STOP of the machine	LED Out	LED turns OFF
SO3 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO4 LED Out	LED ON	LED Out	LED ON
SO5 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO8 LED Out	LED ON	LED Out	LED ON

Tab.: 3.2.I

- Key switch on set-up mode / All guard doors closed / No EMERGENCY STOP activated / Machine stopped
- Each "third-party" EMERGENCY STOP switch should be activated one after the other. The functions should be checked according to the following table.
- If no "third-party" EMERGENCY STOP mechanism is present, the jumpers for the "third-party" EMERGENCY STOP should be removed.
- All the malfunction and help displays on the touch-screen must be checked. (if provided)

Visual inspection			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
SO1 LED Out	LED ON	LED Out	LED ON
SO2 LED Out	LED turns OFF EMERGENCY STOP of the machine	LED Out	LED turns OFF
SO3 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO4 LED Out	LED ON	LED Out	LED ON
SO5 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO8 LED Out	LED ON	LED Out	LED ON

Tab.: 3.2.II

Inputs/outputs which are not used cannot be checked. Which outputs on the respective machine are to be used is to be found in the connection diagram.

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3.3 Test of all the safety switches with and without locking

- Key switch on production mode / All guard doors closed / No EMERGENCY STOP activated / Machine ON (Machine OFF and guard doors unlocked for safety switches with guard locking)
- Each safety switch should be opened or unlocked one after the other. The functions should be checked according to the following table.
- All the malfunction and help displays on the touch-screen must be checked (if provided).

Visual inspection			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
SO1 LED Out	LED ON	LED Out	LED ON
SO2 LED Out	LED ON	LED Out	LED ON
SO3 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO4 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO5 LED Out	LED ON	LED Out	LED ON
SO8 LED Out	LED turns OFF	LED Out	LED turns OFF

Tab.: 3.3.I

- Key switch on set-up mode / All guard doors closed / No EMERGENCY STOP activated / Machine stopped
- Each safety switch should be opened or unlocked one after the other. The functions should be checked according to the following table.
- All the malfunction and help displays on the touch-screen must be checked. (if provided)

Visual inspection			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
SO1 LED Out	LED ON	LED Out	LED ON
SO2 LED Out	LED ON	LED Out	LED ON
SO3 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO4 LED Out	The LED turns OFF after 2.5 seconds delay	LED Out	The LED turns OFF after 2.5 seconds delay
SO5 LED Out	LED ON	LED Out	LED ON
SO8 LED Out	LED turns OFF	LED Out	LED turns OFF

Tab.: 3.3.II

Inputs/outputs which are not used cannot be checked. Which outputs on the respective machine are to be used is to be found in the connection diagram.

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3.4 Test of single-channel interruptions of all sensors

For basic devices connected by double-channel, the opening of a contact (e.g. broken wire) is not immediately detected as a fault. However, the device naturally does turn OFF. If this individual contact is closed again, the basic device shows a malfunction since both contacts did not switch. The malfunction is only eliminated by renewed synchronous switching of both contacts. This behaviour is to be checked for all switching devices with sensors connected by double-channel and potential-free (not possible for OSSD inputs such as e.g. light grids).

In order to diagnose a broken wire, the switching state of a sensor (if signal contact is provided) is compared to the switching state of the corresponding switching device in the standard PLC. The corresponding touch-screen message (if provided) is to be checked.

Initial situation:

- Machine stopped
- Switching device to be checked in turned-on status

Test procedure:

- Interrupt channel 1 by disconnecting terminal S11 (disconnecting S12 when the machine is equipped with non-contact safety switches)
- Check the functions according to the following table.
- All the malfunction and help displays on the touch-screen (if provided) must be checked.

Visual inspection at the individual basic devices and if applicable at their contact extension devices			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
LED "In1"	The LED turns OFF / without delay	LED "In1"	The LED turns OFF / without delay
LED "In2"	LED ON (continuous light)	LED "In2"	LED ON (continuous light)
LED "Out"	The LED turns OFF / without delay	LED "Out"	The LED turns OFF / without delay
LED "Fault"	LED OFF	LED "Fault"	LED OFF

Tab.: 3.4.I

Test procedure:

Close channel 1 by connecting terminal S11 (connecting S12 when the machine is equipped with non-contact safety switches)

Check the functions according to the following table.

Visual inspection at the individual basic devices and if applicable at their contact extension devices			
Safety switching device (basic device)		Contact extension device (if provided)	
Where	What	Where	What
LED "In1"	LED ON (blinking light)	LED "In1"	LED remains OFF
LED "In2"	LED ON (continuous light)	LED "In2"	LED ON (continuous light)
LED "Out"	LED OFF	LED "Out"	LED remains OFF
LED "Fault"	LED ON	LED "Fault"	LED remains OFF

Tab.: 3.4.II

→ The malfunction on the switching device is eliminated by activation of the sensor (synchronous switching of the contacts).

→ to return the machine to operation mode, the reset button must be pressed.



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3.5 Check the EMERGENCY STOP / SAFETY SWITCH overview in the connection diagram against the actual configuration

The connection diagram contains an overview in FBG = .1101 in which the positions of the individual EMERGENCY STOP / SAFETY SWITCH mechanisms are entered. This sheet must be compared with the actual configuration and adapted if necessary!

For changes in the diagram, these should be updated in CAD by the person responsible and the new version number of the connection diagram is to be entered in the test report.

3.6 Testing the EMERGENCY STOP of the machine at maximum machine output

At maximum machine output, activate an EMERGENCY STOP mechanism whilst observing the display of the frequency inverter. No warnings (overcurrent, excessive torque, etc.) may be displayed. In addition, observe whether the machine shuts down smoothly; thereafter, no fault may appear (except for the EMERGENCY STOP).

3.7 Third-party machine EMERGENCY STOP jumper marked with notice

A notice must be attached to the jumpere for EMERGENCY STOP of a third-party machine. Should a third-party machine exchange the EMERGENCY STOP function with our machine, this should be tested as described in item 3.2.

3.8 Determine the delay by observation and enter it in the test report

The delay time should not deviate from the standard unless this was so established by the responsible person in charge (for the delay times, see 1.4). Check that the disconnecting does not take place before the machine standstill, since after the disconnecting the machine merely coasts down slowly.

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5 Test of the safety-related potentials

5.1 Functional test for frequency inverters for dangerous drives

Description:

- The inverters are turned OFF using the "Safe stop" input. The module DriveGuard safe off for Powerflex article no. 20A-DG01, type: relay control: 24V - is required in order to realise the safety function "Safe stop" for PF40P and PF70EC.
- The safe stop is selected (turned OFF) via the safety contact "Combined safety output" (K3).

Note: (terminal designation)

- Danfoss -> 37
- AllenBradley -> Driveguard safe off for Powerflex Extension -> 3
- Acopos -> X1.8 oder X1.9 (linked with each other internally)
- Acopos Multiple shaft 1 -> X1.7 and X1.9
- Acopos Multiple shaft 2 -> X1.1 and X1.3
- SEW -> X17.SVI24
- An indication on the display of the frequency inverter may not be used to determine the input status of the "safe stop" since the indication is not safe.

Test procedure:

- Measure the voltage at terminal (see note above) for the safe stop → 24 volts
- Press an EMERGENCY STOP button
- All the frequency inverters must be in "safe stop" status.
- Measure the voltage at the terminal → 0 volts
- Restore the initial situation
- Measure the voltage at the safe stop terminal → 24 volts
- Open a guard door
- All the frequency inverters must be in "safe stop" status. (Exception - discharge conveyor relay controlled)
- Measure the status at the terminal → 0 volts

Exceptions:

- Energy stores are not turned OFF by means of safe stop

5.2 Functional test of the terminal devices which are subject to a safety-related shutdown

Description:

- Components which are subject to a safety-related shutdown are selected (turned OFF) by means of the safety contacts of the corresponding switching devices (e.g. 13/14, 23/24, 33/34) or, if applicable, of their contact extension devices.
The contacts and their descriptions are to be found in the connection diagram (follow the cross-references of the extension units).

Test:

- It is to be checked whether the connected components or subassemblies are turned OFF after activation of an EMERGENCY STOP switch or the triggering of a protective device.

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5.3 Test of the wiring between the safe outputs, their contact extensions and the terminal devices

Description:

- The wiring of the safe outputs/inputs (corresponding switching devices (e.g. 13/14, 23/24, 33/34) or, if applicable, their contact extension devices) to the components and terminal devices is to be checked according to the connection diagram.

Note:

- If all the tests were already performed in the control cabinet according to MO-E008 or MO-E009 and MO-010, item 5.3 can be answered with "yes". If "not included" is recorded in the test report of the control cabinet, then the tests entered under "Remarks" must still be conducted before the confirmation (test reports are to be found with the connection diagram in the control panel).

Test:

- Activate the EMERGENCY STOP.
- Perform wiring tests between all terminals of the safe outputs or inputs and their contact extensions (see the description in Chapter 1.3) and the terminal devices.

5.4 Functional test of the pneumatic components which are subject to a safety-related shutdown – label carriage forward

Description:

- The selection of "label carriage forward" is effected with a double circuit by means of two outputs and two valves

Note:

- The power supply of the valve manifold is turned OFF with enable circuit 3

Test pneumatic:

- No EMERGENCY STOP activated
- Guards closed
- The valve for "label carriage back" may not be activated
- Both valves must be selected for the function "label carriage forward". A single valve may not trigger the movement of "label carriage forward".
- First valve manually switched there is no reaction from the labeling carriage
- Second valve manually switched there is no reaction from the labeling carriage
- Label carriage forward, contemporaneous manually by activating of both valves → Label carriage moves forward
- Label carriage back by manually activating → Label carriage moves back

Test electric:

- No EMERGENCY STOP activated
- Guards closed
- The valve for "label carriage back" may not be activated → disconnect valve
- Bridge relay contact for "label carriage forward" → Label carriage moves forward
- Reset Lamp Emergency, Guard and SLS is off.
- Eliminate bridge → Reset Lamp is on → label carriage is forward
- Connect Valve label carriage back → Label carriage moves backward

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6 Safely-limited speed (SLS)

6.1 Hardware

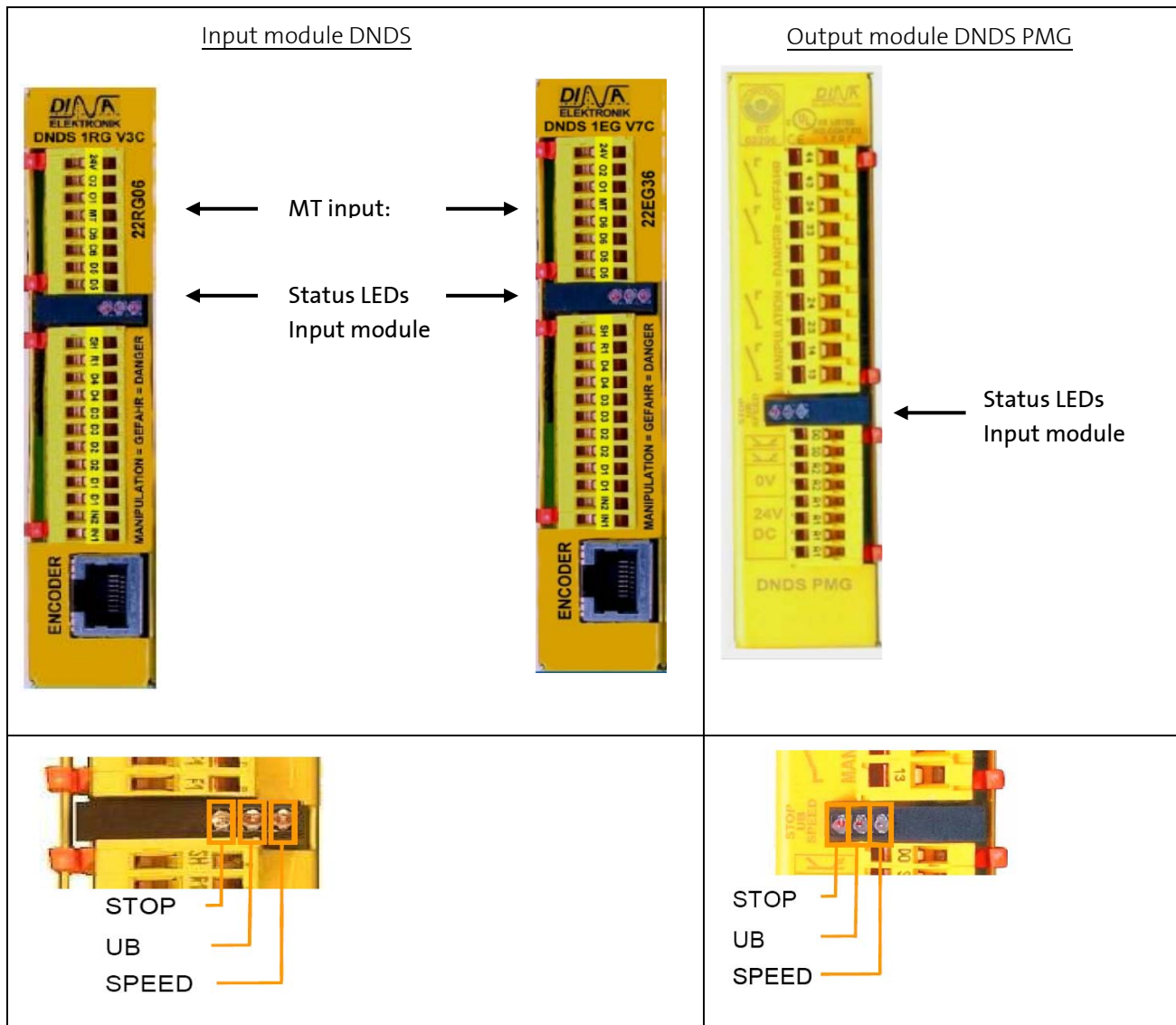


Fig.: 6.1.1

6.2 Description SLS

6.2.1 Function

- The DINA device monitors all safety-relevant motors with regard to exceeding speed limits.
- The monitoring is only effected in set-up mode with opened safety device (low level at the "MT" inputs). If the DINA device reports a malfunction, the malfunction is read via an ASI Safety module and a "safe stop" at the inverters is activated.

6.2.2 RESET

- A RESET is effected via the general RESET button. The voltage supply to the DINA monitoring modules is interrupted by means of a relay.

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6.3 Testing

All DINA monitoring modules are to be tested one after the other. The limit speeds are set by the DIP switches according to the connection diagram.



Danger

Make sure that all drives are in free-wheel status (if necessary remove components, e.g. starwheels).

Test (for each drive)

- Key switch on set-up mode
- Guard doors open, 24 volts at the MT input falling off, DINA analysis device is active
- In the event of more than one DINA device, those devices that are not being tested must be supplied with 24 volts at the MT terminal, so that they are not active.
- Increase the JOG speed to approx. 500 containers/h greater than the limit speed.
- → JOG the machine → The machine must stop
- The "speed LEDs" of the tested input module and of the output module must have turned OFF.
- RESET
- → All the "speed LEDs" are lit up again
- Set the JOG speed to the correct value again.
- In the event of more than one DINA device, remove the 24 volt power supply again and connect the original wiring.

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7 Options

Every change or deviation from the work instructions is to be documented in the checklist.
The respective test procedure is to be described.

Tests which were not performed are also to be documented here in this chapter.

8 Changes to the safety technology

8.1 Any deviations to the safety technology should be documented in the following (if present)

There are two empty fields in the test report, one is for the documentation of any deviations of the safety technology from the standard and the other is for the documentation of changes in the safety technology after completed acceptance inspection (see the test report).

8.2 Changes to the safety technology after the acceptance inspection has been completed

If a change is performed in the safety technology after the acceptance inspection, the entire acceptance inspection must not be repeated if it can be assumed that an individual test suffices for the change. To record what was changed, please write a brief description of what was changed and checked!

8.3 Description of the change by the design engineer

All the changes must have been prescribed by the design engineer. The sequential process for changes is described in the following document. (look Intranet -> Areas » MO » QM Documents » Work Instructions + Test Reports » QM in general » QM Work Instruction MO007)

8.4 Change tested by commissioning engineer

All functions affected by the change must be checked.

8.5 Pack a copy of the completed test report in transparent film and attach it to the connection diagram at the front

It is important that the report is filled out completely because it is the basis for the on-site tests.

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9 Test report workflow

The test reports for the respective Work Instructions are saved in the Intranet (Areas » MO » QM Documents » Work Instructions + Test Reports). They must be placed in the assembly folder of the machine. The person in charge must fill in the actual values. The assembly documents with the test reports must be handed over to the subsequent workstation as assembly progresses. When final assembly is complete, the test reports are to be deposited as originals in the "blue folder". A copy of the test report is filed at the front in the electrical connection plan folder and accompanies the machine.

Prüfprotokoll MO-IE T012				
Geltungsbereich		Montage IET		
Anwendungsbereich		Inbetriebnahme Etikettier- und Inspektionstechnik		
Titel des Prüfprotokolls				
Kommissions-Nr.:		SAP-Auftrag:		
Auftragnehmer:		Anlagen-Nr.:		
Kunde:		Land:		
Regionalbetreuer:		KRONES Baustellenleiter:		
Zu beachtende Arbeitsunterlagen:				
Prüfpunkte: ☐ n. V. = nicht vorhandener Prüfpunkt ☐ i. O. = in Ordnung/Prüfpunkt erfüllt				
Prüfpunkte				
1.	Beschreibung Description	I. O.	n. V.	Unterschrift
1.1.	Arbeitsanweisung wurde durchgelesen. Alle Prüfungen werden nach AA durchgeführt. The work instructions have been read through. All tests will be performed according to the work instructions.	☐	☐	
	Versionsnummer Arbeitsanweisung: Version of work instructions:	☐	☐	
2.	Funktionsprüfung ASI Safety Kombigerät Functional test of the ASI Safety combination unit	I. O.	n. V.	Unterschrift
2.1	Prüfung K1 mit Kontakterweiterung durchgeführt Test K1 with contact extension performed	☐	☐	
2.2	Prüfung K2 mit Kontakterweiterung durchgeführt Test K2 with contact extension performed	☐	☐	
2.3	Prüfung K3 mit Kontakterweiterung durchgeführt Test K3 with contact extension performed	☐	☐	
2.4	Prüfung K4 mit Kontakterweiterung durchgeführt Test K4 with contact extension performed	☐	☐	
2.5	Prüfung K5 mit Kontakterweiterung durchgeführt Test K5 with contact extension performed	☐	☐	
2.6	Prüfung K6 mit Kontakterweiterung durchgeführt Test K6 with contact extension performed	☐	☐	
2.7	Prüfung K7 mit Kontakterweiterung durchgeführt Test K7 with contact extension performed	☐	☐	
2.8	Prüfung K8 mit Kontakterweiterung durchgeführt Test K8 with contact extension performed	☐	☐	
2.9	Prüfung K9 mit Kontakterweiterung durchgeführt Test K9 with contact extension performed	☐	☐	
2.10	Prüfung K10 mit Kontakterweiterung durchgeführt Test K10 with contact extension performed	☐	☐	

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Fig.: 9.1

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10 Tests at the destination by the on-site assembly team

10.1 Tests at the destination by the on-site assembly team

During the commissioning of the machine at the destination, all safety functions that could not be tested at the factory must be tested. This also includes tests that basically belong to the in-house assembly tests insofar as they were not marked as tested in the test report.

10.2 Test of the position of the EMERGENCY STOP / SAFETY SWITCHES which are disassembled after acceptance for shipping

Based on the respective diagram in the connection diagrams the proper assembly of the individual EMERGENCY STOP / SAFETY SWITCHES which were disassembled is to be checked on site. The object designation at the switch must agree with the diagram in the connection diagrams.

10.3 Processing of open items from the previous chapters

All tests from the previous chapters not marked with "yes" in the test report must be performed.